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BCAC591 — PHP with MySQL | Semester V

PROJECT DETAILS REPORT

1. Project Title

Campus Resolve — Student Complaint Management System

2. Problem Statement

Students regularly encounter facility, infrastructure, and laboratory issues across campus — such as malfunctioning classroom fans, defective lab workstations, drinking water cooler breakdowns, and damaged experiment equipment — but there is currently no organized, transparent way to lodge grievances or track their resolution. Verbal or paper complaints lead to lost records, lack of administrative accountability, and zero visibility for students regarding when repairs will occur. Campus Resolve provides a centralized, role-based platform where students can submit categorized complaints with precise campus locations and track progress through a transparent audit timeline, while campus administrators prioritize issues, record timestamped remarks, manage status transitions, and resolve grievances with verifiable accountability.

3. User Roles

The system has two roles, matching the minimum requirement set in the project notice:

- **Student (Complainant)** — Registers with verified student credentials and academic department; logs in to access a personal workspace dashboard; views personal complaint metrics and resolution rates; submits new complaints with category, campus location, description, and optional image attachment; tracks real-time status and historical timeline of their own complaints; and manages personal profile details.
- **Admin (Campus Operations / Authority)** — Logs in separately via dedicated administrator authentication; oversees campus operations through an analytics dashboard (reviewing pending queues, active tickets, registered student count, and 7-day activity volume); reviews all submitted grievances across departments; updates ticket statuses (*Pending → In Progress → Resolved / Rejected*) with official remarks recorded in the audit history; and monitors student accounts.

4. Module List — Mapped to This Project

Every mandatory module from the project notice, mapped specifically to this project's subject matter:

#	Mandatory Module	How It Applies to This Project
1	**Authentication**	Student registration with validation (name, email, department dropdown, password match/strength) and login; separate Administrator login. Passwords securely hashed using `password_hash()`.
2	**Role-Based Access**	Two roles — Student and Administrator — each with a dedicated dashboard, distinct permission barriers, and session checks (`require_student()` and `require_admin()`).
3	**Dashboard**	Student dashboard displays personal complaint metrics (Total, Pending, In Progress, Resolved) and resolution rate %. Admin dashboard shows campus-wide counts, active triage queue, and 7-day activity chart.
4	**Core Entity CRUD**	The core entity is **Complaint** (supported by **Complaint Updates**). Students Create and View complaints. Admins View, Update status/remarks, and manage the full grievance lifecycle.
5	**Listing Page**	Complaints listing view — student personal complaints table and admin campus-wide registry with Search (by complaint code/title) and Filter (by category, department, and status).
6	**Detail View**	Clicking a complaint opens a comprehensive detail view displaying complaint code, student info, department, location, description, attached photo, and full chronological timeline of admin remarks.
7	**Status Workflow**	Each complaint moves through 4 states: *Pending → In Progress → Resolved* (or *Rejected*). Each transition creates a timestamped record in `complaint_updates`. Only Admin can update status.

#	Mandatory Module	How It Applies to This Project
8	**Admin Panel**	A separate Admin area providing centralized visibility over all complaints across departments, with action forms to assign remarks, advance status, and monitor student accounts.
9	**Form Validation**	Complaint submission form validates title, category whitelist, location, description length, and attachment format; registration validates email uniqueness, department list, and password complexity in PHP, not just HTML5.
10	**Flash Messages**	Session-based flash message system (`set_flash()`, `get_flash()`) displaying alert banners on actions: complaint logged, status transition saved, profile updated, login failure, etc.
11	**Profile Management**	Students and Admins can view and edit their own name, contact email, assigned department, and change account passwords.

Optional modules included, since they genuinely apply to this project:

- **Chronological Audit Timeline / Event History Tracking** — Every status transition and administrative remark generates an immutable audit record in `complaint_updates`, giving both students and administrators complete, accountable transparency.
- **Campus Operations Analytics & Visual Trends** — Admin dashboard calculates real-time resolution rates, displays status breakdown metrics, and plots past 7-day complaint activity frequency.
- **File Handling & Evidence Attachment** — Students can attach image/document evidence of damaged infrastructure or equipment when submitting a complaint; files are validated, securely stored server-side in `uploads/complaints/`, and previewed in the detail view.

5. Database Tables (Schema)

The following is the database schema implemented in MySQL using InnoDB engine with foreign key constraints and UTF-8 encoding.

Table: users

Column	Type / Notes
id	INT, Primary Key, Auto Increment
user_code	VARCHAR(20), Unique — e.g. ST-2016, AD-001
name	VARCHAR(100), Not Null
email	VARCHAR(120), Unique, Not Null
password	VARCHAR(255), Not Null — stored using password_hash()
role	ENUM('student', 'admin'), Not Null, Default 'student'
department	VARCHAR(100), Default Null — academic department affiliation
status	ENUM('active', 'inactive'), Not Null, Default 'active'
created_at	DATETIME, Not Null, Default CURRENT_TIMESTAMP

Table: complaints

Column	Type / Notes
id	INT, Primary Key, Auto Increment
complaint_code	VARCHAR(20), Unique — e.g. CR-1042
user_id	INT, Foreign Key → users.id (ON DELETE CASCADE)
title	VARCHAR(200), Not Null — short summary of grievance
category	ENUM('Infrastructure', 'IT', 'Facilities', 'Laboratory', 'Other'), Not Null
location	VARCHAR(150), Not Null — e.g. Block B Room 204, Central Library
description	TEXT, Not Null — comprehensive issue description
attachment	VARCHAR(255), Nullable — server storage path of uploaded evidence
status	ENUM('Pending', 'In Progress', 'Resolved', 'Rejected'), Default 'Pending'
created_at	DATETIME, Not Null, Default CURRENT_TIMESTAMP
updated_at	DATETIME, Not Null, Default CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Table: complaint_updates

Column	Type / Notes
id	INT, Primary Key, Auto Increment
complaint_id	INT, Foreign Key → complaints.id (ON DELETE CASCADE)
admin_id	INT, Nullable, Foreign Key → users.id (ON DELETE SET NULL)
status	VARCHAR(50), Not Null — status state (Submitted, Under Review, In Progress, Resolved, Rejected)
remark	TEXT, Not Null — official administrative remark or resolution notes
created_at	DATETIME, Not Null, Default CURRENT_TIMESTAMP

6. Page List

The following table outlines all system interfaces, their intended audience, primary function, and current implementation status in the codebase:

#	Page / Screen	Accessible By	Purpose	Status
1	**Landing Page (`index.php`)**	Public	System introduction, workflow showcase, and role-based entry links	IMPLEMENTED
2	**Student Sign In (`auth/login.php`)**	Public	Student authentication with email, password verification, and CSRF check	IMPLEMENTED
3	**Admin Sign In (`auth/admin_login.php`)**	Public	Dedicated administrative portal sign in with admin role validation	IMPLEMENTED
4	**Register (`auth/register.php`)**	Public	Account registration with department dropdown and password validation	IMPLEMENTED
5	**Sign Out (`auth/logout.php`)**	Student / Admin	Terminates session, deletes cookie, and redirects to login	IMPLEMENTED
6	**Student Dashboard (`student/dashboard.php`)**	Student	Personal metrics (Total, Pending, In Progress, Resolved), rate %, recent tickets	IMPLEMENTED
7	**Admin Dashboard (`admin/dashboard.php`)**	Admin	Campus-wide KPI counters, active triage queue, 7-day activity chart	IMPLEMENTED
8	**Submit Complaint (`student/submit_complaint.php`)**	Student	Grievance submission form with category, location, and attachment upload	PARTIALLY IMPL.
9	**My Complaints (`student/complaints.php`)**	Student	Searchable, filterable list of student's own submitted complaints	PLANNED
10	**Complaint Detail (`student/complaint_details.php`)**	Student / Admin	Complete issue details and chronological timeline of administrative remarks	PLANNED
11	**Student Profile (`student/profile.php`)**	Student	View student ID, department info, and update password / contact info	PARTIALLY IMPL.
12	**All Complaints (`admin/complaints/list.php`)**	Admin	Master campus grievances listing with category, department & status filters	PLANNED
13	**Admin Review (`admin/complaints/detail.php`)**	Admin	Inspect complaint details, review attached evidence, and enter remarks	PLANNED
14	**Update Status (`admin/complaints/update_status.php`)**	Admin	Backend handler saving status transition and logging timeline record	PLANNED
15	**User Management (`admin/users.php`)**	Admin	Directory of registered students across departments with status controls	PLANNED
16	**Admin Profile (`admin/profile.php`)**	Admin	View administrator account details and manage credentials	PARTIALLY IMPL.

7. File Structure

The project repository follows a modular, role-separated directory layout adhering to clean MVC-like separation of concerns. Below is the tree of main, relevant application files:

```
campus-resolve/
│
├── config/
│   └── db_connect.php            # MySQL database connection using PDO (prepared statements, utf8mb4)
│
├── includes/
│   ├── auth.php                  # Session management, role guards (require_student, require_admin), CSRF
│   ├── helpers.php               # HTML escaping, status badges, date formatters, departments list
│   ├── auth_check_student.php    # Guard middleware for student area
│   ├── auth_check_admin.php      # Guard middleware for admin area
│   ├── header.php                # Common top navigation bar
│   ├── footer.php                # Common document footer
│   ├── student-sidebar.php       # Navigation sidebar for student workspace
│   └── admin-sidebar.php         # Navigation sidebar for admin workspace
│
├── assets/
│   ├── css/
│   │   └── styles.css            # Central stylesheet (design tokens, components, responsive layout)
│   └── js/
│       └── app.js                # Client-side validation, UI interactions, and mobile menu toggle
│
├── auth/
│   ├── register.php              # Student account registration with input validation
│   ├── login.php                 # Student portal sign in
│   ├── admin_login.php           # Dedicated administrator sign in
│   └── logout.php                # Session destruction and sign out
│
├── student/
│   ├── dashboard.php             # Student landing: KPI cards, resolution rate %, recent 5 complaints
│   ├── submit_complaint.php      # Grievance submission form with evidence upload
│   ├── complaints.php            # [Planned] Personal complaints listing and status filter
│   ├── complaint_details.php     # [Planned] Detail view with chronological update timeline
│   └── profile.php               # Student profile and contact settings
│
├── admin/
│   ├── dashboard.php             # Operations dashboard: campus KPIs, 7-day chart, active queue
│   ├── complaints/
│   │   ├── list.php              # [Planned] Master campus complaints table with search & filters
│   │   ├── detail.php            # [Planned] Admin complaint inspection & remark entry
│   │   └── update_status.php     # [Planned] POST handler for status transitions & update timeline
│   ├── users.php                 # [Planned] Directory of registered students across departments
│   └── profile.php               # Administrator profile management
│
├── database/
│   ├── schema.sql                # DDL: users, complaints, complaint_updates tables
│   └── seed.sql                  # Initial dataset: default admin, test students & sample complaints
│
├── uploads/
│   └── complaints/               # [Planned] Storage directory for uploaded complaint evidence
│
└── index.php                     # Public landing page: system overview and role entry
```